```
library IEEE;

use IEEE.STD_LOGIC_1164.ALL;


entity PmodKYPD_tb is
end PmodKYPD_tb;

architecture Behavioral of PmodKYPD_tb is

    component PmodKYPD
        Port (
            clk : in STD_LOGIC;
            JA  : inout STD_LOGIC_VECTOR(7 downto 0);
            an  : out STD_LOGIC_VECTOR(3 downto 0);
            seg : out STD_LOGIC_VECTOR(6 downto 0)
        );
    end component;

    signal clk : STD_LOGIC := '0';
    signal JA  : STD_LOGIC_VECTOR(7 downto 0);
    signal an  : STD_LOGIC_VECTOR(3 downto 0);
    signal seg : STD_LOGIC_VECTOR(6 downto 0);

    signal key_row : STD_LOGIC_VECTOR(3 downto 0) := "0111";
    signal key_col : STD_LOGIC_VECTOR(3 downto 0) := "0111";
```

```
signal row_drive : STD_LOGIC_VECTOR(3 downto 0);

type slv4_array is array (0 to 15)
    of STD_LOGIC_VECTOR(3 downto 0);
type slv7_array is array (0 to 15)
    of STD_LOGIC_VECTOR(6 downto 0);

--Keypad row positions
constant rows : slv4_array := (
    "0111", "1011", "1101", "1110",
    "0111", "1011", "1101", "1110",
    "0111", "1011", "1101", "1110",
    "0111", "1011", "1101", "1110"
);

--Keypad column positions
constant cols : slv4_array := (
    "0111", "0111", "0111", "0111",
    "1011", "1011", "1011", "1011",
    "1101", "1101", "1101", "1101",
    "1110", "1110", "1110", "1110"
);
```

```
--Expected seven segment outputs
constant expected_seg: slv7_array := (
    "1111001", --1
    "0011001", --4
    "1111000", --7
    "1000000", --0
    "0100100", --2
    "0010010", --5
    "0000000", --8
    "0001110", --F
    "0110000", --3
    "0000010", --6
    "0010000", --9
    "0000110", --E
    "0001000", --A
    "0000011", --B
    "1000110", --C
    "0100001"  --D
    );

begin

    --DUT
```

```
uut: PmodKYPD
    port map (
        clk => clk,
        JA  => JA,
        an  => an,
        seg => seg
    );

-- Clock generation (100 MHz)
clk_process: process
begin
    clk <= '0';
    wait for 5 ns;

    clk <= '1';
    wait for 5 ns;
end process;

-- Connect keypad rows
JA(7 downto 4) <= row_drive;

-- Do not drive keypad column outputs
JA(3 downto 0) <= (others => 'Z');
```

```
-- Keypad simulation
process(JA, key_row, key_col)
begin
    row_drive <= "1111";
    if JA(3 downto 0) = key_col then
        row_drive <= key_row;
    end if;
end process;

-- Test all 16 keys
stim_proc: process
begin
    report "Starting Keypad Test";

    for i in 0 to 15 loop
        key_row <= rows(i);
        key_col <= cols(i);
        -- Wait for keypad scanning
        wait for 4.1 ms;

        -- Check seven segment output
```

```
            assert seg = expected_seg(i)
                report "Keypad output mismatch"
                severity error;

            assert an = "1110"
                report "Anode output mismatch"
                severity error;

        report "Key test passed";

    end loop;

    report "ALL 16 KEYS TESTED SUCCESSFULLY";
        wait;

    end process;

end Behavioral;
```